

Tools	Pros	Cons
Ansible	1. SSH-based, so it doesn't require the installation of any agents on remote nodes. 2. Playbook structure is simple and clearly structured.	1. Less powerful than tools based in other programming languages. 2. Variable registration is required for even basic functionality, which can make easier tasks more complicated.
Chef	1. Extendable with scripting language, easy configuration files 2. Easy to implement, learn and use on daily basis for configuration management	1. Web UI does not allow modification of configurations 2. Other platforms have better pre-configured deployment scripts
Docker	1. Docker produces an API for container management 2. Fast application deployment	1. Containers don't work at bare-metal rates. 2. Data storage is intricate
Puppet	1. Well-established support community through Puppet Labs. 2. Simple installation and initial setup	1. For more advanced tasks, you will need to use the CLI (Command Line Interface), which is Ruby based. 2. Because of the DSL (Domain Specific Language) and a design that does not focus on simplicity, the Puppet code base can grow large and unwieldy.
Kubernetes	1. Microservices rolling updates 2. Makes it a lot easier to establish effective CI/CD pipelines.	1. Businesses need a certain degree of reorganization when using Kubernetes with an existing app. 2. Pods sometime need a manual start/restart before they start working as intended. This can happen in certain situations such as when running near full capacity

Figure 6: Pros and cons of the five tools featured

Ansible	Chef	Docker	Kubernetes	Puppet
Configuration management tool.	Configuration management tool.	It's a container technology.	Configuration management tool.	Configuration management tool.
It is written in Python.	Ruby and Erlang.	Written in Go programming language.	Written in Go programming language.	Ruby-DSL (domain-specific language) language.
Easy for configuration management	Complex from the development perspective	Easiest to manage, understand and isolate.	Setting up Kubernetes requires a lot of planning	Difficult for beginners
It is more appropriate for front-end developers, where some programming might be needed.	It's similar to Puppet	It is very different from the others and has several components	Kubernetes is suitable for developers of modern applications	It's more targeted towards operations that don't require development background
It is similar to Puppet in producing files.	Builds a pipeline of processes.	It delivers configuration for one process at a time, making Docker files simpler than bash script for process configuration.	Kubernetes brings a lot of complexity as it is a larger project with more moving parts than any other DevOps tool, but that also leads to latency in executing commands, making troubleshooting and monitoring cumbersome.	It may configure more than one process at a time and make the dependencies a bit complex.

Figure 7: Comparing the top five DevOps tools

takes containerisation to the next level. It functions admirably with Docker or any of its options. Kubernetes is still exceptionally new; its first version turned up in 2015. It was established by a few Google engineers who were looking for a solution to oversee

containers at scale. With Kubernetes, you can gather your containers into legitimate units.

Key features

Some of the platform features that Kubernetes offers are:

- Container grouping using Pod
- Self-healing
- Auto-scalability
- DNS management
- Load balancing
- Rolling updates or rollbacks
- Resource monitoring and logging

Figure 6 depicts the pros and cons of the five tools featured here. Figure 7 compares these five top DevOps automation tools.

The road ahead

The DevOps universe is brimming with unique and remarkable open source tools. Mainstream DevOps tools can help you overcome any hindrance. You can select the device that suits your business needs and immediately watch the improvements unfold in your business activities. Diverse DevOps tools will not only work well independently, but also work well in combination. **END** 🐧

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